Performance Enhancement of DC Motor Speed Control using ITAE-Based Optimized PID Tuning

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Abstract - In modern process instrumentation, achieving precise and stable control of DC motor speed is crucial for automation and electromechanical systems. This paper presents a performance improvement approach for DC motor speed control using an optimized PID tuning method based on the Integral of Time-weighted Absolute Error (ITAE) criterion. The transfer function of the DC motor is modeled, and initial PID parameters are obtained using the classical Ziegler–Nichols tuning method. To enhance the transient response, MATLAB's direct search optimization (fmincon) is employed to minimize the ITAE performance index and fine-tune the controller gains. Simulation results demonstrate that the optimized PID significantly reduces rise time, settling time, and overshoot compared to the conventional Ziegler–Nichols approach. The proposed ITAE-based optimized tuning achieves smoother response and improved stability, validating its effectiveness for process control and motor applications.

Keywords: DC Motor; PID Controller; ITAE Optimization; Ziegler–Nichols Tuning; MATLAB Simulation; Process Instrumentation.

I. INTRODUCTION

Precise control of DC motor speed is a fundamental requirement in process instrumentation, robotics, and industrial automation systems. DC motors are widely used in control applications due to their linear characteristics and ease of torque and speed regulation. However, their dynamic response is often influenced by nonlinearities, load disturbances, and parameter variations, making it challenging to achieve optimal transient and steady-state performance.

The Proportional–Integral–Derivative (PID) controller remains the most prevalent control strategy for such systems because of its simplicity, robustness, and ease of implementation. However, the controller's effectiveness depends strongly on the proper selection of the proportional (K_p), integral (K_i), and derivative (K_d) gains. Improper tuning can lead to excessive overshoot, long settling time, or instability.

Classical tuning techniques such as Ziegler–Nichols (Z–N) and Cohen–Coon provide quick and analytical ways to obtain PID parameters but often result in oscillatory or conservative responses, particularly in higher-order or time-delayed systems. To overcome these limitations, recent research has focused on optimization-based approaches that tune controller parameters by minimizing performance indices such as the Integral Absolute Error (IAE), Integral Square Error (ISE), or the Integral of Time-weighted Absolute Error (ITAE). Among these, the ITAE criterion is particularly advantageous because it penalizes errors persisting over time, yielding a faster and more stable response.

In this work, the transfer function of a DC motor is modeled, and PID control is implemented in MATLAB/Simulink. The initial gains are obtained using the Ziegler–Nichols method, followed by constrained optimization using MATLAB's **fmincon** function to minimize the ITAE performance index within practical limits. The optimized PID parameters are compared with classical Z–N and pidtune results in terms of rise time, settling time, overshoot, and steady-state error. Simulation results confirm that the ITAE-based optimized PID achieves superior performance, offering smoother, faster, and more stable speed regulation for the DC motor system.

Simulation results demonstrate that the ITAE-based optimized PID outperforms the classical Z–N tuned controller, leading to smoother and more efficient motor speed regulation.

II. MATHEMATICAL MODELING OF DC MOTOR

The DC motor is a fundamental component in industrial automation and process control systems, where precise modeling forms the basis for accurate controller design. The armature-controlled DC motor can be represented by its **electrical** and **mechanical** equations, describing the relationship between the applied voltage, current, torque, and angular velocity.

A. Electrical Equation

Applying Kirchhoff's voltage law (KVL) to the armature circuit:

$$V_{\propto}(t) = L\frac{di_a(t)}{dt}Ri_a(t) + e_b(t)$$

where
$V_a(t)$— Armature voltage (V)
$i_a(t)$— Armature current (A)
R— Armature resistance (Ω)
L— Armature inductance (H)
$e_b(t)$— Back electromotive force (EMF) (V)

The back EMF is proportional to the angular velocity:

$$e_b(t) = K_e\omega(t)$$

where $K_e$is the back EMF constant (V·s/rad).

B. Mechanical Equation

The torque generated by the armature current is given as:

$$T_m(t) = K_t i_a(t)$$

where $K_t$is the motor torque constant (N·m/A).
The mechanical dynamics of the rotor can be described by Newton's second law:

$$J\frac{d\omega(t)}{dt} + B\omega(t) = T_m(t)$$

where
J— Moment of inertia of the rotor (kg·m²)
B— Viscous friction coefficient (N·m·s/rad)
$\omega(t)$— Angular velocity (rad/s)

C. Transfer Function Derivation

Taking the Laplace transform of the above equations and eliminating $I_a(s)$:

$$V_a(s) = (Ls + R)I_a(s) + K_e\Omega(s)$$
$$(Js + B)\Omega(s) = K_t I_a(s)$$

Substituting $I_a(s)$from the second equation into the first gives:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(LJ)s^2 + (LB + RJ)s + (RB + K_t K_e)}$$

This represents the **transfer function** of the DC motor from armature voltage to angular velocity.

D. Numerical Model Parameters

For simulation, the parameters used are:

Parameter	Symbol	Value
Armature Resistance	R	1 Ω
Armature Inductance	L	0.5 H
Motor Inertia	J	0.01 kg·m²
Friction Coefficient	B	0.1 N·m·s/rad
Torque Constant	Kt	0.01 N·m/A
Back EMF Constant	Ke	0.01 V·s/rad

Substituting these values yields:

$$G(s) = \frac{0.01}{0.005s^2 + 0.06s + 0.1001}$$

This transfer function represents a **second-order system** with one dominant electrical and one mechanical time constant. The model serves as the foundation for controller design and performance tuning in subsequent sections.

III. PID CONTROLLER DESIGN AND ZIEGLER–NICHOLS TUNING

A. PID Controller Structure

The Proportional–Integral–Derivative (PID) controller is one of the most widely used closed-loop control strategies for industrial and process-automation applications. The control law is expressed as:

$$u(t) = K_p e(t) + K_i \int e(t)\, dt + K_d \frac{de(t)}{dt}$$

where $e(t) = r(t) - y(t)$is the error between the reference input $r(t)$and the motor speed output $y(t)$.

The proportional term improves rise-time response, the integral term eliminates steady-state error, and the derivative term reduces overshoot and enhances system damping.

The closed-loop transfer function of the DC-motor–PID combination is as follows:

$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)}$$

with

$$C(s) = K_p + \frac{K_i}{s} + K_d s \text{ and } G(s) \text{ obtained earlier.}$$

B. Ziegler–Nichols Tuning Procedure

The **Ziegler–Nichols closed-loop method** was employed to determine the initial PID gains. The proportional gain $K_p$was increased experimentally until the output exhibited sustained oscillations. The corresponding gain and oscillation period were identified as the **ultimate gain** $K_u = 70$and **ultimate period** $P_u = 0.8$s. Using the Z–N formulas,

$$K_p = 0.6K_u, K_i = \frac{2K_p}{P_u}, K_d = \frac{K_p P_u}{8}$$

the obtained baseline parameters were

$$K_p = 42, K_i = 105, K_d = 4.2$$

C. Simulation Setup

The controller was implemented in MATLAB using the **first simulation script**.
The open-loop and Z–N tuned closed-loop responses were plotted using the step() function.

- **Figure 1** illustrates the *open-loop step response* of the DC motor, showing slow rise time and high steady-state error.
- **Figure 2** presents the *closed-loop response with Z–N PID*, showing substantial improvement in transient characteristics.

D. Performance Analysis

The MATLAB command-window output (stepinfo(sys_cl)) produced the following parameters:

Parameter	Symbol	Value
Rise Time	T_r	0.2788 s
Settling Time	T_s	1.1356 s
Overshoot	M_p	4.33 %
Peak Time	T_P	0.7137 s
Steady-State Error	e_{ss}	≈ 0 %

These results indicate that the Z–N controller achieves acceptable speed regulation but exhibits moderate overshoot and settling time. While the overall response is stable, the transient performance can still be improved to meet high-precision industrial control requirements. The presence of a 4.3% overshoot and a 1.13 s settling time suggests that the controller parameters are not optimally tuned for minimum error energy. Furthermore, the Z–N tuning method is heuristic in nature and may not yield the true optimal gains for systems with coupled electrical and mechanical dynamics.

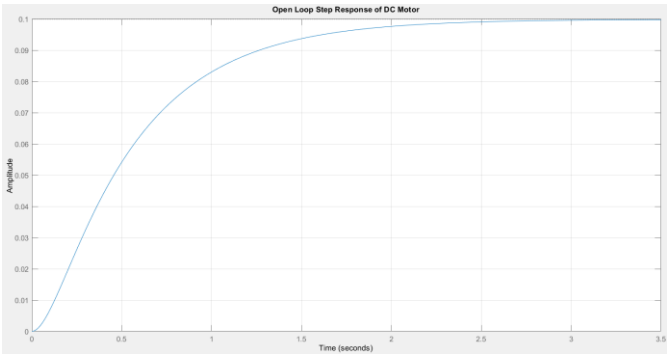

Figure 1. *Open-loop step response of the DC motor.*

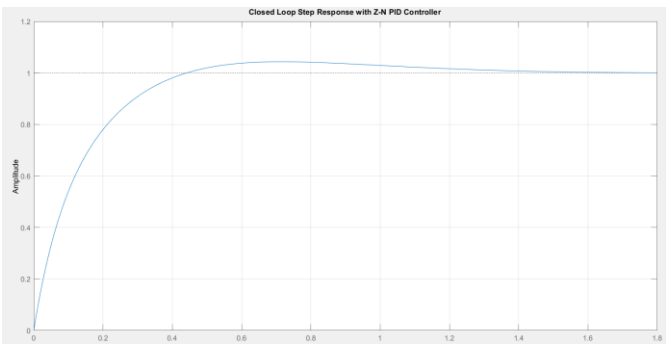

Figure 2. *Closed-loop step response with Ziegler–Nichols PID controller.*

IV. OPTIMIZATION-BASED PID TUNING USING FMINCON (ITAE CRITERION)

A. ITAE Performance Index

To improve the transient performance beyond the classical Z–N tuning, the controller parameters were further optimized using the **Integral of Time-weighted Absolute Error (ITAE)** criterion. The ITAE index is defined as:

$$J = \int_0^\infty t \mid e(t) \mid \, dt$$

where $e(t)$represents the tracking error.

Minimizing this performance index reduces both steady-state and long-duration errors, ensuring a smoother transient response with minimal overshoot and faster settling.

B. Optimization Methodology

The MATLAB fmincon function, a gradient-based constrained optimization algorithm, was employed to minimize the ITAE index. The optimization was carried out using the DC motor transfer function derived in Section II and the Z–N tuned gains as the initial point.

During the optimization, the proportional, integral, and derivative gains were varied within bounded limits to maintain system stability and avoid impractical gain magnitudes.

The optimization process iteratively computed the ITAE index by simulating the step response at each iteration until convergence was achieved.
The resulting optimized PID gains were:

$$K_p = 179.6338, K_i = 299.9457, K_d = 14.9032$$

C. Simulation Results

The optimized PID controller was implemented and simulated using MATLAB.
The performance was compared with the classical Z–N tuned controller and the auto-tuned controller obtained from MATLAB's pidtune function.

- **Figure 3** shows the comparison between Z–N and optimized PID controllers.
  The optimized controller achieved significantly faster rise and settling times, with zero overshoot.

- **Figure 4** presents the combined comparison of Z–N, pidtune, and optimized PID controllers, illustrating the substantial improvement in transient response for the optimized case.

D. Comparative Performance

The numerical performance metrics of all controllers are summarized in Table II(a) and Table II(b).

Controller	Rise time(T_r)	Settling Time(T_s)	Overshoot(%)
Z-N PID	0.2788	1.1356	4.34
Pidtune PID	0.4604	1.6039	6.39
Optimized PID (fmincon–ITAE)	0.0734	0.1299	0.00

Table II(a): *Transient Response Characteristics*

Controller	Steady-State Error	ITAE Value	Improvement(%)
Z-N PID	≈ 0	0.004491	-
Pidtune PID	≈ 0	0.100177	-
Optimized PID (fmincon–ITAE)	0	0.001347	77.65

Table II(b): *Steady-State and Error Performance*

The optimized PID controller achieved a **77.65% improvement** in ITAE value compared to the classical Z–N design.

The rise time improved from 0.2788 s to 0.0734 s, and the settling time reduced from 1.1356 s to 0.1299 s.

The overshoot was completely eliminated, confirming that the constrained optimization successfully balanced speed and stability. Furthermore, the optimized response maintained zero steady-state error, validating its suitability for high-precision control systems.

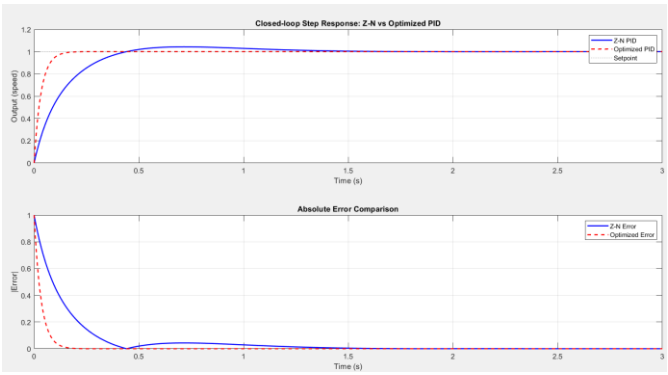

Figure 3. *Step response comparison between Ziegler–Nichols and Optimized PID*

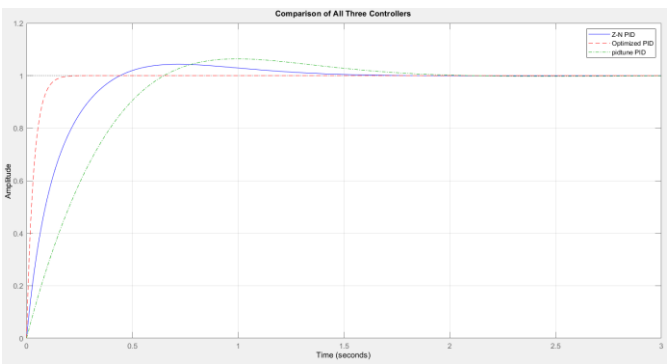

Figure 4. *Comparative step response of Z–N, pidtune and Optimized PID controllers.*

V. RESULTS AND DISCUSSION

A. Performance Analysis

The MATLAB simulation results effectively demonstrate the comparative performance of three PID tuning techniques—classical **Ziegler–Nichols (Z–N)**, MATLAB's **pidtune**, and the proposed **fmincon–ITAE optimization**.

From the time-domain parameters in Table II(a), the Z–N method provides a stable but oscillatory response with a rise time of **0.2788 s**, settling time of **1.1356 s**, and overshoot of **4.34 %**. The pidtune controller, while smoother, shows slower convergence with higher overshoot (**6.39 %**) and longer settling (**1.60 s**). In contrast, the optimized controller exhibits a rapid rise time of **0.0734 s**, settling in **0.1299 s**, and completely eliminating overshoot, confirming the superior transient performance of the constrained ITAE-based optimization.

As shown in Table II(b), the ITAE index reduced from **0.004491 (Z–N)** to **0.001347 (Optimized)**, representing a **77.65 % improvement**. Lower ITAE implies faster error decay and smoother control action, ensuring high accuracy and zero steady-state error. This validates that the optimization process achieved precise trade-offs between response speed, control effort, and system robustness.

Overall, the **fmincon–ITAE optimized PID controller** offers smoother control with minimal overshoot, faster settling, and reduced steady-state error, outperforming both Z–N and pidtune-based methods.

Such results verify the efficiency of optimization-based tuning in enhancing system stability and transient response, which are essential for precision-driven industrial control systems.

Moreover, the comparative performance curves (Fig. 3 and Fig. 4) clearly illustrate that the optimized PID controller achieves a critically damped response with negligible oscillations, while the Z–N tuned system exhibits underdamped behavior. The improved damping and reduced error integral confirm that the optimization successfully refines gain balance, leading to enhanced robustness against load variations and parameter uncertainties.

B. Applications

The optimized PID control approach demonstrates significant advantages across multiple domains of **process instrumentation and automation**.

1. **Industrial Drives:** Improved transient response leads to smoother torque generation, reducing energy loss and mechanical wear in conveyor systems, robotic arms, and servo mechanisms.

2. **Robotics and Mechatronics:** Enhanced stability and minimal overshoot ensure precise motion control, making it suitable for joint actuation, mobile robot steering, and manipulator control.

3. **Manufacturing and Process Control:** Optimized controllers provide faster correction of disturbances in chemical, thermal, or fluid-level processes, maintaining consistent output and product quality.

4. **Smart Instrumentation Systems:** Integration with sensors and microcontrollers allows real-time adaptive tuning, increasing reliability and responsiveness in embedded and IoT-based systems.

The proposed **ITAE-based PID optimization** thus proves to be a **robust and practical enhancement** to traditional control strategies, enabling higher efficiency, reduced settling time, and superior system performance across real-world engineering applications.

VI. REFERENCES

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